

Murphy-Health

An Estimation-Style Iterative Benchmark for Longevity LLMs

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The Problem

Current longevity LLM benchmarks (incl. *LongeBench*) have three blind spots that let a model look smart without actually *being calibrated*:

- **Limited topic coverage.** RNA-level aging biology — enzymes, disease/hallmark associations, pathway regulation — is sparsely represented relative to its weight in real aging research.
- **Detection only, no estimation.** Tasks are dominated by binary / multi-class classification. Few measure a model's ability to put a *calibrated number* on an aging biomarker.
- **One-shot, no self-correction.** Existing benchmarks score a single final answer. None test whether a model can *iterate* under partial feedback or *recognize its own mistakes* when its guess is rejected.

Real biology demands calibrated uncertainty *and* the ability to refine — right answers for the wrong reasons should not pass.

The Solution



$$S = \left(10 + \sum_{i \in \mathcal{G}} \left\lfloor \frac{u_i}{l_i} \right\rfloor\right) \cdot 2^{N-|\mathcal{G}|}$$

$[l_i, u_i]$ = last interval on problem i ;
 $\mathcal{G}$ = set of GOOD final answers; lower S wins.



Anchors the scientific basis of our new question bank.



1. RNA Question Bank

What we built. Two curated RNA-aging question banks published openly on HuggingFace: *carolw/EP04* (enzyme–disease pairs from aging-phenotype enzymes) and *sarahliu/rna-eb0x* (RNA expression, hallmark and pathway regulation).

How we built it. Seeded from primary literature, auto-labeled with abstract retrieval, hand-verified, then *adversarially filtered* against SotA LLMs — only questions where strong models disagree or err survive.

Why it matters. Plugs the RNA / early-detection gap in LongeBench, and is *ungated*: one CLI flag pulls it.

2. The Algo — Estimation

Mechanic. Each problem gets an interval $[l_i, u_i]$. The model sees *only* GOOD / BAD — no directional hint. A shared budget of $\lfloor \frac{18}{3}N \rfloor$ slips is split across all N problems; cap 3 slips/problem.

Why this works. Width factor $\lfloor u/l \rfloor$ punishes hedged “ $[1, 100]$ ” answers; the $2^{N-|\mathcal{G}|}$ term punishes leaving problems empty. Refinement is a *voluntary bet*: random guessing wins 50%, so beating 50% is genuine biological reasoning, not bracket-search.

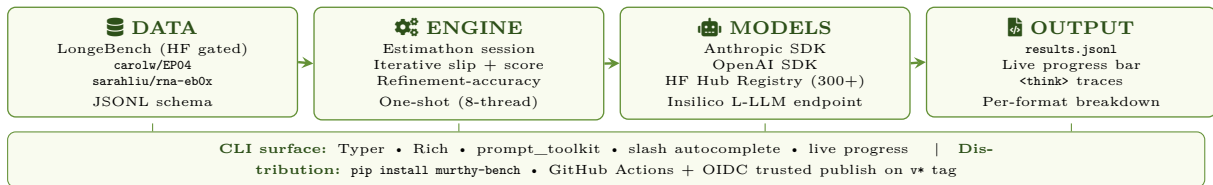
3. The Program — CLI

Install. `pip install murthy-bench`; first run launches an API-key + HF-token wizard.

One command, any model, any data. `murthy run -model X -provider Y -tasks Z` works for Anthropic, OpenAI, the Insilico L-LLM endpoint, or any HF inference URL — mix-and-match.

Interactive chat. `/test`, `/batch`, `/model`, `/explore` slash commands with autocomplete; live progress bar, per-slip `<think>` trace capture, JSONL results for downstream analysis.

Pipeline



Team



Yiqi (Michael) Yao
Harvey Mudd College, '29
Joint CS & Mathematics

miyao@g.hmc.edu

CLI architecture, scoring
engine, packaging & test
pipeline



Yan (Olivia) Yan
Harvey Mudd College, '29
Math & Computational Biology

oyan@g.hmc.edu

Insilico L-LLM integration,
LongeBench wiring,
open-source model curation
for benchmark testing, head
chef 🍳



Yuhui (Sarah) Liu
Harvey Mudd College, '28
Joint Chemistry & Biology

saraliu@g.hmc.edu

RNA research lead —
disease/hallmark, pathway
regulation; Q-bank curation
& SotA testing



Xiaokuan (Carol) Wang
Harvey Mudd College, '29
Math & Computational Biology

carowang@g.hmc.edu

RNA research —
aging-phenotype enzymes;
Q-bank curation & SotA
testing



Chun-Yu (Alex) Chen
Claremont McKenna College, '29
Math & Computational Biology

cchen29@cmc.edu

Project architecture,
scoring-function ideation,
documentation, diversity
debugging, two-day
round-trip driver 🚗